ANSWERABILITY AWARE EVALUATION OF HALLUCINATION AND RELIABILITY IN RETRIEVAL AUGMENTED GENERATION FOR TECHNICAL DOCUMENTATION QUESTION ANSWERING

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List of Abbreviations

Slno.	Abbreviation	Full Form
1	RAG	Retrieval Augmented Generation
2	LLM	Large Language Model
3	QA	Question Answering
4	BM25	Best Matching 25 lexical ranking function
5	DPR	Dense Passage Retrieval
6	AUROC	Area Under the Receiver Operating Characteristic Curve
7	F1	Harmonic mean of precision and recall
8	ECE	Expected Calibration Error
9	NLI	Natural Language Inference
10	API	Application Programming Interface

Abstract

Retrieval-Augmented Generation is currently implemented in numerous question-answering systems when an answer needs to be generated from technical documentation. Such an approach is effective since the system retrieves relevant documents and uses them to generate answers. However, the problem with retrieval in RAG models is that even when the system finds a relevant document, there is no guarantee that the context is sufficient. Indeed, sometimes retrieved chunks might be insufficient, irrelevant, outdated, and yet, the system generates a response with high confidence. Such an approach can lead to false guidelines and waste engineers' time.

Therefore, this work investigates the problem of answerability-aware RAG model assessment for technical documentation QA. Answerability-aware models evaluate whether the retrieved context is sufficient before making decisions about providing responses. The primary data source for this research will be the TechQA RAG evaluation dataset, while RAGTruth will be a supplementary source for analyzing hallucinations. The following techniques will be utilized for retrieving documents, including document cleaning and splitting, lexical retrieval, and dense retrieval. Once the context is obtained, the system computes answerability features based on which it trains a classifier that predicts whether the question can be answered based on context.

After that, the system answers or requests additional evidence, depending on the answerability confidence score. The final result will be assessed with retrieval recall, answerability performance, unsafe answer rate, false abstention rate, unsupported claim rate, ROUGE-L and BERTScore when available, and error analysis. This research does not intend to provide a solution for preventing hallucinations. Instead, its focus will be on finding the boundary where RAG model should answer and abstain from responding.

Key words: Retrieval Augmented Generation, Technical Documentation Question Answering, Answerability, Hallucination Evaluation, Context Sufficiency, Abstention, Reliability, Hybrid Retrieval, RAGTruth, TechQA RAG Eval

1. Background

Nowadays, there is an increased importance of technical documentation question answering because of the increasing popularity of technical question answering assistants among users of software systems, cloud computing platforms, and enterprise tools. It is no longer necessary for users to read extensive documentation in order to find solutions to even the smallest problems; rather, users would prefer to ask a concise question and receive an equally concise answer.

The reason why the TechQA dataset could be valuable for research into technical QA tasks lies in the fact that it is based on practical technical support situations, thus being suitable for research into technical QA, where the proper understanding of the technical context plays a role (Castelli *et al.*, 2020).

Earlier question answering benchmarks such as MS MARCO and Natural Questions helped move QA research closer to real user queries and web-based evidence (Nguyen *et al.*, 2016; Kwiatkowski *et al.*, 2019). SQuAD 2.0 also made an important point by adding unanswerable questions, where the correct behavior is to know that the passage does not contain enough information (Rajpurkar, Jia and Liang, 2018). This is very close to the situation in technical documentation. A user can ask a valid question but the retrieved document may still not contain the answer.

Retrieval Augmented Generation tries to solve part of this problem by retrieving external documents before generating an answer. The RAG approach by (Lewis *et al.*, 2020) showed how retrieval and generation can be combined for knowledge intensive tasks. Dense Passage Retrieval by (Karpukhin *et al.*, 2020) improved open domain retrieval using dense vector representations, while models such as Fusion in Decoder used multiple retrieved passages during generation (Izacard and Grave, 2021) . These ideas are useful for technical QA because they make the model less dependent on its internal memory.

However, retrieval itself does not make the answer reliable. Even if RAG retrieves some document chunks, but chunks may be incomplete, noisy, outdated, noisy or from different old version. Model will still generate an answer which looks correct but it will not be grounded with the retrieved context.

And this is the core issue in traditional RAG. Confidence level will be high while evidence won't be reliable. In technical documents, a wrong answer will lead to wrong configuration, wrong debugging procedure, and inaccurate knowledge about a product.

The latest studies that evaluate the quality of RAG models are following suit too. Some examples include ARES, RAGAs and RAGChecker, as they not only check the response generated by RAG models but also check some other aspects of the response (Saad-Falcon *et al.*, 2024). The RAGTruth tool has a dataset of manually annotated RAG output hallucinations, which can help identify false or contradicting statements (Niu *et al.*, 2024). Finally, the most recent work focused on determining whether the retrieved context is sufficient (Joren *et al.*, 2025) .

This research proposal addresses the problems mentioned above and others. But unlike previous studies, which only checked whether the response provided was valid, this study first considers whether the system needs to provide a response based on its assessment whether there is enough evidence. This study emphasizes answerability-aware evaluation, wherein the system evaluates the strength of evidence before answering.

2. Literature Review

I will review the existing work in three area that is related to this research proposal: technical document question answering, retrieval augmented generation and hallucination or answerability evaluation.

2.1 Technical Documentation and Document Question Answering

The TechQA dataset by (Castelli *et al.*, 2020) provides an interesting basis for this work since it concerns technical support question answering. In contrast to typical QA datasets that include generic questions, the TechQA dataset comprises technical questions asked in the forum environment. As a result, the TechQA dataset will be relevant in this research because it includes questions concerning product's behavior, errors, configuration, and troubleshooting.

MS MARCO by (Nguyen *et al.*, 2016) has shown the importance of dealing with imperfect retrieved evidence. Similarly, the authors of the Natural Questions dataset by (Kwiatkowski *et al.*, 2019) have used Google searches and their corresponding snippets as data. Consequently, both datasets represent useful sources of background for the investigation, which connects QA to the problems of evidence retrieval and imperfect user intents.

The SQuAD 2.0 dataset by (Rajpurkar, Jia and Liang, 2018) has introduced the notion of answerability, which means that QA models should learn to detect whether the evidence contains an answer to the user's question. Thus, a technical RAG model should have similar capabilities and avoid providing answers in case if the retrieved context is insufficient.

In addition, the problems of question answering over long documents and grounded question answering have been addressed in several datasets: QASPER – QA over long documents, e.g. research papers (Dasigi *et al.*, 2021) . KILT – knowledge-intensive task learning (Petroni *et al.*, 2021) . Moreover, document grounding and evidence selection have been examined in the QuAC and MultiDoc2Dial datasets that include QA in an information-seeking dialogue scenario (Choi *et al.*, 2018; Feng *et al.*, 2021) .

2.2 Retrieval Augmented Generation

A Retrieval Augmented Generation (RAG) was introduced by (Lewis *et al.*, 2020) to integrate the parametric language model with non-parametric external knowledge sources. This method is highly effective for such system and domain having a large amount of knowledge sources. For the same reason, it is also effective for the technical documentation domain because the information in the technical documentation keeps changing. Rather than re-training the model with such information, it becomes efficient to use this information with RAG approach to answer the questions. The REALM (Guu *et al.*, 2020) , and DPR (Karpukhin *et al.*, 2020) contributed significantly towards recognizing the importance of retrieval in language model-based QA tasks. Especially, the DPR study is quite important for this research proposal as it demonstrates the role of dense embedding in retrieving passages on the basis of semantic similarities rather than simple exact matches. In this context, it is also important to note that the technical documentation includes exact errors codes and product names which need to be answered lexically through BM25 approaches. The FID, or Fusion in Decoder by (Izacard and Grave, 2021) shows that the retrieved passages can be incorporated into a generative model for answering open-domain questions. The ColBERT (Khattab and Zaharia, 2020) also adds value to this study in that it shows improvement in neural retrieval by integrating late interaction between query and passage tokens. These studies are highly relevant to the hybrid retrieval approach used in this research proposal.

Self RAG study by (Asai *et al.*, 2024). explains how the model can perform retrieval, generation, and judgment of its own answer. The self-generation capability of the model is quite interesting but, in this research, a simpler solution is adopted by introducing a decision boundary. The model will make decisions about answering, requesting evidence or abstinence based on answerability confidence.

2.3 Hallucination, Factuality and Answerability Evaluation

Hallucination in large language models has been an identified problem. For instance, TruthfulQA by (Lin, Hilton and Evans, 2022) showed that the model might provide inaccurate responses despite sounding reasonable. SelfCheckGPT by (Manakul, Liusie and Gales, 2023) proposed black-box methods for detecting hallucinations through consistency across sampled answers. FActScore by (Min *et al.*, 2023) shifted the focus of evaluation to small factual claims. The method is crucial in cases when the response is rather lengthy and thus its whole answer cannot be considered incorrect. Only partial sentence may be wrong if a part of it in the answer was consistent with the retrieved context while another sentence might lack support from retrieved context. Therefore, it is necessary to investigate hallucination at the claim level. For RAG-based systems, hallucination and retrieved context should be investigated together. RAGTruth (Niu *et al.*, 2024) is related directly to the proposal because it involves annotation of hallucinations within RAG responses both in terms of overall answer and word level. Thus, it becomes clear that retrieval hallucination cannot be solved with retrieval, as Model can give unsupported information.

Some frameworks have been developed to assess RAG in terms of hallucination systematically. RAGAs evaluate retrieval and generation based on such metrics as faithfulness and answer relevance (Es *et al.*, 2024). ARES assesses context relevance, answer faithfulness and answer relevance using lightweight judges based on synthetic data (Saad-Falcon *et al.*, 2024). Lastly, RAGChecker presents a detailed analysis of retrieval and generation phases (Ru *et al.*, 2024). They all are important as they demonstrate that RAG systems require evaluation as pipelines and not just by looking at their answers.

The most relevant works in this research area would include Sufficient Context by (Joren *et al.*, 2025). which investigates the question of whether there is enough context for answering and explores guided abstention. The approach resembles the main concept behind this proposal, as it can be applied to technical documentation QA, where the need to abstain rather than generate a fluent response plays an essential role.

Another work that is relevant in this case is Selective Question Answering under Domain Shift by (Kamath, Jia and Liang, 2020). The research investigates and presents the details of when a QA system should abstain, especially when confident prediction becomes impossible due to the deviation from the training distribution. The paper provides support for the use of calibrated answerability scores and decision thresholds in the proposed framework.

2.4 Research Gap

Today, most researches in QA are dedicated to providing an accurate response to the posed question. Researches in RAG are also dedicated to answering the posed question.

Researches dedicated to the QA for technical documentation are scarce. When using a regular RAG model, all technical documents will be ingested into the system, after which the model will provide an accurate response, even when the information in the document does not match the required information; therefore, the information provided by the model may be inaccurate or irrelevant, insufficient or outdated.

The purpose of my research is to fill this gap with a new approach to answerability boundary, ensuring that the ingestion of technical documents will retain their original meaning, the retrieved context will be sufficient or not, the user has the right to receive an answer to the given question, or the model will avoid responding.

In my research, I will focus on investigating answerability boundaries with classifications of answers, decision-making policies, and risks of hallucinations.

3. Research Questions (If any)

- How frequently do the answers generated by a RAG system on technical documentation questions turn out to be non-answerable because of insufficient retrieved context?
- What are the characteristics of retrieval and contexts useful in determining answerability on technical questions?
- What is the effect of the answerability-aware policy on the unsafe answer rate and false abstention rate?
- How is the risk of hallucination measured without overclaiming in the process?

4. Aim and Objectives

In this research, I propose to construct and test an answerability-aware Retrieval-Augmented Generation system for question answering in technical documents. This work aims at evaluating whether the model will generate an answer, ask for additional proof, or refrain from providing a response because the context is insufficiently.

Objectives:

- Building a hybrid retrieval pipeline based on lexical retrieval and dense embedding.
- Take advantage of retrieval confidence, lexical overlap, dense score, score gap, and context-related features to train the answerability classifier.
- Implement a decision-making policy for answering, asking for further information, and abstaining from answering.
- Evaluate the proposed model by considering reliability-based evaluation measures like unsafe answer rate, false abstention rate, and unsupported claims rate.

5. Significance of the Study

This research work has great importance because the technical document assistant tool can serve many purposes in many sectors. In case the system replies with inaccurate answers, it can mislead and confuse the user. The technical documentation assistant tool can be utilized by the user for requesting API, configurations of the system on cloud, internal database behavior analysis, setup, and configuring system properties. Inaccurate answer of the query may affect the work of the user, cause operational error, impact the score in the benchmarking process and consume lot of time.

The research will focus on examining if the system has enough data to answer the query, the user has the access to view the answer to the query or in case the system does not have adequate data to reply the query then it should reply that it doesn't have enough data to reply the query from the documents retrieved.

This research would benefit researchers who conduct RAG evaluation tests, developers involved in building technical documentation question answering system, teams who make use of the RAG based assistant tool for their support work and the end users using the technical document assistant tool.

6. Scope of the Study

This study centers on answerability aware assessment of Retrieval Augmented Generation on technical documentation QA. Specifically, it seeks to determine whether the retrieval is sufficient enough to generate an answer before the generation process kicks off. The scope provides information about the areas that fall under the study, as well as its limitation areas.

6.1 Scope of the Research

In the current research I will focus on technical documentation question answering based on RAG. As the main source for data analysis, I will use TechQA RAG Eval dataset that has technical questions, answers and contexts to be used in evaluation (NVIDIA, 2024). As an auxiliary dataset for evaluation of hallucination issues, I will use RAGTruth dataset (ParticleMedia, 2024). Data cleaning, construction of documentation corpus, chunking, hybrid retrieval, extraction of answerability features, answerability classification, calibration, decision making using thresholding, generation and evaluation of answers will be included in this research. The system will produce one of the following actions: providing an answer, requesting additional evidence or abstaining.

The inclusion of automatic evaluation metrics and manual error analysis in the current work will be made. The latter step is taken since the automatic scores can be insufficient to fully understand why there was a problem, whether it was caused by incorrect retrieval of facts, lack of proper context, generation process or decision thresholding.

6.2 Out of Scope for this Study

- This research does not consider training a new model from scratch as well as developing an entire system for deployment. Additionally, complete hallucination detection is not considered in this study.
- Only technical documentation QA is considered in this study.

7. Research Methodology

The approach that will be adopted in this experiment involves the use of evaluation. The answerability aware RAG model pipeline will be used, where technical documentations available in the dataset TechQA RAG Eval (NVIDIA, 2024) will be cleaned, tokenized, indexed with retrieval based on lexical as well as dense embeddings retrieval. The keyword based retrieval will be performed through BM25. Chunks thus retrieved will be used for the computation of answerability features, which are then used to train a classifier for the estimation of whether the retrieved context is adequate to answer the question asked by the user.

7.1 Data Collection

The main dataset that will be employed in this research will be collected from two public sources. The main dataset will be TechQA RAG Eval from NVIDIA on Hugging Face (https://huggingface.co/datasets/nvidia/TechQA-RAG-Eval) by (NVIDIA, 2024). This dataset is suitable because it is aligned with technical documentation question answering and also contains examples where the question, answerability status, answer and context can be studied together. RAGTruth (https://github.com/ParticleMedia/RAGTruth) will be used as a supporting resource for hallucination analysis (Niu *et al.*, 2024; ParticleMedia, 2024) because it provides annotated RAG outputs with hallucination labels.

7.2 Data Preprocessing and Corpus Construction

The dataset will first be converted into a tabular structure. The question, answer, answerability label and context fields will be cleaned and normalised. The documentation corpus will be built from the available contexts in TechQA RAG Eval (NVIDIA, 2024) . Long contexts will be split into overlapping chunks so that retrieval can return the most relevant part of the document instead of only using the beginning of a long passage.

The corpus construction will be handled carefully. This document corpus is treated as the available technical documentation collection for the RAG system. However, the train, validation and test splits will be kept separate for model training, threshold tuning and final evaluation.

7.3 Dataset Splitting

For an effective training process as well as evaluation process, the dataset needs to be divided into three parts through stratified sampling. This will ensure that the class is evenly distributed in each of the subsets. Stratified sampling is preferred because this will distribute answerable and non-answerable questions effectively in all three subsets. Training dataset will be used to train the answerability classifier, while validation dataset will be used for threshold tuning, and testing dataset will be used only for evaluation purposes.

Subset	**Percentage**	**Purpose**
Training set	70%	Used to train the answerability classifier
Validation set	15%	Used to tune calibration and answerability decision thresholds
Test set	15%	Used for final evaluation of retrieval, classifier and decision policy

Table 7.3.1 Dataset split for training, validation and testing

7.4 Hybrid Retrieval

For technical questions, QA requires both exact match and semantic match. Retrieval would utilize both lexical retrieval and dense retrieval. The purpose of using lexical retrieval is to retrieve those exact names of products, exact parameters of them, as well as exact error codes and configuration information, since there will be fixed values in technical documentation that would get distorted with the use of semantic matching. Sentence embeddings from Sentence Transformer will help to find semantic similarity between query and document text.

7.5 Answerability Feature Extraction

After the documents are retrieved, features will then be extracted from both the question and the retrieved document(s). The following features will be included: BM25 score, dense score, hybrid score, the difference in score between top ranked contexts, lexical matching, number of retrieved contexts, length of the context, length of the question.

7.6 Answerability Classifier and Calibration

The classifier used for this part of the model would be a Random Forest Classifier, which will predict whether the question can be answered based on the retrieved information. This would be done using the retrieval and context-based features obtained from the above processes. Calibration by isotonic regression needs to be done due to poor calibration in raw probability estimates.

7.7 Decision Policy

Calibrated answerability score would be used to determine system's actions in such a way that high score leads to answering, medium score – to requesting for more evidence and low score – to abstaining. Thresholds for answerability score would be tuned on the validation set through balance between rate of unsafe answers and false abstentions. This is the key answerability-aware part of our model.

7.8 Answer Generation

For those situations in which the action selected is to respond, a sequence-to-sequence language model will be applied to construct the response based on the extracted context. When generating answers, this study shall adopt FLAN T5 models due to their instruction fine-tuning capabilities and that they can be executed on a Jupyter notebook framework (Chung and Hou, 2024) . When the generated answer is poor in quality or appears to be an outright refusal, extraction from the retrieved context could be done as an alternative. The point is not to cover up the failure to generate; it is to stay close to the source documents.

7.9 Hallucination Risk and Unsupported Claim Analysis

Unsupportable claims will be detected based on a proxy for unsupported claim estimation. The answers produced by the models will be divided into subclaims and matched with context segments using embeddings. If the subclaims do not have strong evidence to support them, then such subclaims will be considered unsupported. This will not be considered an absolute hallucination detection system but rather a flag for unverifiable answers. RAGTruth (ParticleMedia, 2024) will be used to identify patterns of hallucinations.

7.10 Evaluation Metrics

Evaluation will take place at the stages of retrieval, answerability classification, and the final decision. Recall at k will measure the quality of the retrieval process. Accuracy, precision, recall, F1 score, and AUROC will evaluate the performance of the answerability classifier. Decision accuracy, percentage of unsafe answers, false abstentions, answer coverage, unsupported claims, ROUGE L, and BERTScore will evaluate the system's final performance.

Unsafe answer rate = unanswerable questions answered by the system / total unanswerable questions

False abstention rate = answerable questions not answered by the system / total answerable questions

F1 Score = 2 x Precision x Recall / (Precision + Recall)

7.11 Manual Error Analysis

The manual error analysis will be performed using the test data. The errors will be classified based on different scenarios such as right response, dangerous response, incorrect non-response, wrong retrieval, need more evidence and ungrounded response.

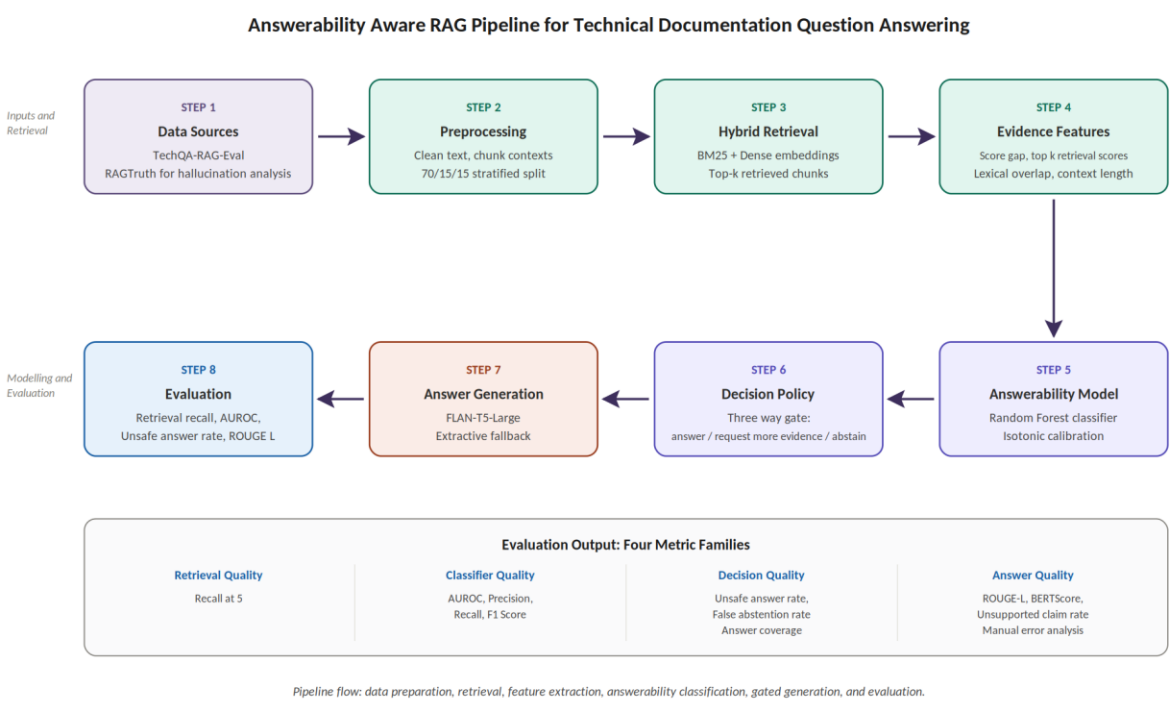

Figure 7.11.1 Proposed Answerability Aware RAG Methodology.

8. Requirements Resources

Hardware & software prerequisites needed for conducting the research are listed below. This project can be developed on a personal computer, and when the need arises for GPU, it could be run on a cloud notebook environment.

8.1 Hardware Requirements

System Details	**Configuration**
Local Machine	Windows 11, 32 GB RAM
Cloud Platform	Kaggle Notebook or Google Colab
GPU	NVIDIA T4 (16 GB VRAM)
Storage	At least 10 GB free space for datasets, embeddings and outputs

Table 8.1.1 Development environment and machine configuration

Initial coding, pre-processing, and testing are done on Windows 11 laptop with 32 GB RAM. However, complete document processing, chunking, creation of embeddings, generation of answers and evaluation studies will be done on Kaggle/Colab since both platforms offer GPU access.

8.2 Software Requirements

SL.n	Software or Library	Purpose
1	Python 3.10 or above	Primary programming language
2	Jupyter Notebook	Development and experimentation
3	Hugging Face datasets	Loading the research datasets
4	Transformers	Answer generation model
5	Sentence Transformers	Dense embedding generation
6	rank bm25 or TF IDF	Lexical retrieval
7	Scikit learn	Classifier training, calibration and metrics
8	Pandas and NumPy	Data processing and analysis
9	ROUGE Score	Text similarity evaluation
10	BERTScore	Semantic answer quality evaluation when available
11	Matplotlib	Charts and result visualization
12	Mendeley Cite	Citation and bibliography management

Table 8.2.1 Language and libraries to be used

9. Research Plan

This research will follow a structure timeline with well-defined phases, addressing key tasks, potential risks and contingency strategies to ensure timely and successful completion.

9.1 Phase 1: Topic finalization and literature review (March to April)

In this process, extensive literature review was carried out. It includes examining available research methodologies, finding any gaps in the existing research and determining how useful RAG would be in answering Technical Question & Answer. Following literature review, three topics were selected and associated with appropriate data set access confirmations and literature sources. The major deliverables of this phase would be the selection and documentation of three research topics alongside their dataset source.

Risk: Research topics may be deemed unsuitable.

Contingency Plan: To counter this risk, three additional topics have been identified beforehand.

9.2 Phase 2: Research proposal submission (April to June)

At this stage, we will concentrate on laying out the key elements of our study such as the problem statement, objectives of our research, key research questions, and methodology. Our primary output at this stage is submitting the final version of the research proposal document.

Risk: Weak research proposal due to lack of clear objectives.

Contingency Plan: To make sure that we submit our research proposal within the set timeframe, we will set up internal deadlines during the drafting of the research proposal. We will submit the initial draft of the proposal by May 15. We will incorporate the feedback of our supervisor by May 25 and submit the final draft before June 1.

9.3 Phase 3: Dataset preparation and retrieval implementation (June 2 – July 13)

This stage will include implementing the proposed methodology into action. TechQA RAG Eval (NVIDIA, 2024) dataset will undergo pre-processing and Eval by performing document corpus preparation, context chunking, and retrieval recall for both BM25 and dense retrieval.

Risk: Chunking may fail to identify the proper context of the technical documents during preprocessing.

Contingency Plan: To handle chunk size problems various sizes and chunk overlap configurations will be utilized. Recall evaluation at top k as well as manual verification of chunks will evaluate the retrieval performance. Kaggle will also be used as an alternative to Google Colab in case of internet issues.

9.4 Phase 4: Answerability classifier and decision policy (July)

This phase involves features extraction, classifier training to classify answerable questions, isotonic calibration, and threshold tuning.

Risk: The classifier might be overly stringent and might choose to abstain even though the answer is there in the retrieval context.

Contingency Plan: Instead of manually choosing the thresholds, I would tune them based on the validation set and select the one which provides a good compromise between safety and effectiveness.

9.5 Phase 5: Final evaluation, analysis and report writing (August to September)

Stage 4 involves full testing evaluation, ablation experiments, hallucination risk assessment, manual error evaluation, and generation of report along with a presentation video.

Risks: The last minute runs can be slow or the results can differ upon reruns.

Contingency Plan: Save output, fix random seeds, note the settings used and begin writing methodology and literature review ahead of time.

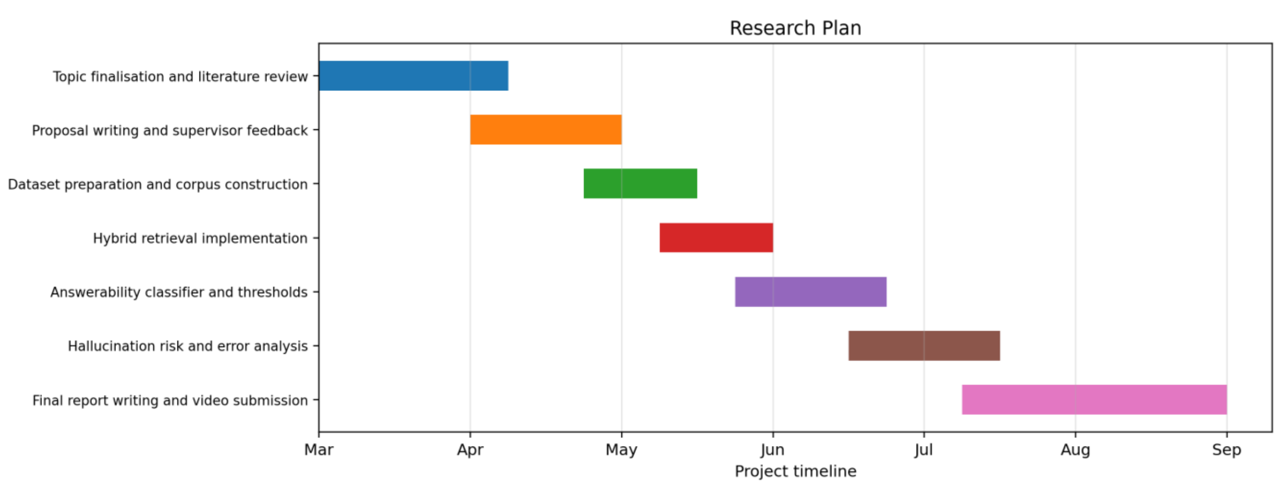

Figure 9.5.1 Research Plan.

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